Spin Ice in the Simplicial Frame: The Ice Rule as Zero Sum, the Lattice Curl as the Cross Product at $e^{i\theta}$, and Kagome Ice as the Tropical Bridge

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Abstract

Spin ice places Ising moments on the bonds of the diamond lattice, along the four tetrahedral directions, which are the $N = 4$ simplicial frame. Its ice rule ("two in, two out") says that the four spins of every tetrahedron sum to zero. We show that in frame coordinates the theory of spin ice reduces to one vector and one matrix from the earlier papers.

The tetrahedron. $\mathbb{R}^4 = H \oplus \mathbb{R}\mathbf{1}$ is the decomposition dipole $\oplus$ monopole: the frame synthesis map gives the tetrahedron's magnetic moment and kills exactly the gauge line, which carries the magnetic charge. The six ice states are the three perfect matchings of K_4 with two orientations each, with moments along the six cubic directions. The nearest-neighbour ice energy is $2Jq^2 - 2J = 6J - \frac{3J}{2}|m|^2$, so ice states are the points of $\{\pm 1\}^4$ farthest from the origin in the genesis metric. Pauling's estimate is the fraction $3/8$ of cube vertices lying in H.

The lattice curl is the cross product at $u = e^{i\theta}$. On the pyrochlore lattice the Bloch curl around the four hexagonal plaquettes is, row by row up to a unit phase, the cross-product generator of Paper 01 evaluated at the exponentiated phase vector: $C(\theta) = \Lambda(\theta)\,\widetilde{K}(e^{i\theta})$. Hence curl grad $= 0$ is Paper 01's $K(u)u = 0$, and the long-wavelength limit is Paper 01b's curl $= \widetilde{K}(\nabla)$. A complexified Paper 01 identity then gives everything else in closed form through the single vector $u_H = e^{i\theta} - \frac{f}{4}\mathbf{1}$, where $f = \sum_j e^{i\theta_j}$ is the diamond band function of Paper 01e:

- *Classical Coulomb phase:* the large-$\mathcal{N}$ spin correlator is $P = I - \frac{1}{4}J - u_H u_H^\dagger/|u_H|^2 = C^\dagger C/(16 - |f|^2)$, with pinch points exactly at the reciprocal lattice.
- *Neutron scattering:* the polarization factor is the same transverse projector evaluated at Q, so the structure factor is a product of two transverse projectors, one for the neutron and one for the ice.
- *Quantum spin ice:* the emergent photon has $\omega(k) = \sqrt{UK}\,\sqrt{16 - |f(k)|^2}$, doubly degenerate, reproducing Benton–Sikora–Shannon's dispersion. Their plaquette vectors $h_{nm} \propto e_n \times e_m$ are the opposite-edge Hodge duals of Paper 01b, Thm. 10.1. By the Lagrange identity, $(\omega/\sqrt{UK})^2 + |f|^2 = 16$: at every k the photon and the diamond electron are the H- and $\mathbf{1}$-components of one Bloch vector. The photon's band top lies exactly on diamond's nodal network.

Kagome ice is the tropical bridge of Paper 01e, realized. In a [111] field each kagome triangle sits at a vertex of the simplex (a dimer). The in-plane field enters as a zero-sum frame vector of Zeeman energies. The free energy is the Ronkin function, so the entropy is $m(1 + x + y)/4 = \frac{3\sqrt{3}}{16\pi} L(\chi_{-3}, 2) = 0.0807664868$ per spin. The in-plane magnetization is the synthesis of the dimer-density vector, which is the gradient of an equivariant gauge functional: it moves from the barycentre (zero field) to a vertex (frozen), exactly the zero-sum and standard-form gauges of Paper 01e. The dimer densities are the turning angles of graphene's Dirac point divided by π. The Kasteleyn transition is the amoeba boundary of Paper 01e, Thm. 8.1, at $T_K = \delta/\ln 2$ for a symmetric tilt. In a [100] field the saturated state is one oriented matching, and the three-dimensional Kasteleyn temperature $2\mu h/(\sqrt{3}\ln 2)$ follows from the matching's two-way branching.

Dipolar spin ice: the long-range interaction is pure gauge plus a short-range residual. The real materials interact through long-range dipolar forces. We prove an exact decomposition: the dipolar energy equals the Coulomb energy of the tetrahedron charges (the gauge components) plus an on-site charge term plus a residual that decays as r^{-5}. All nearest-neighbour physics, including exchange J, is confined to the gauge line. The ice manifold is therefore split only by the residual, which needs no Ewald sum. Its ice-sector bandwidth is $0.178\,D$, 3.7% of the cost of a monopole. Its minimum lies exactly at the X point, independent of J. The selected states are the staggered ice states of the two matchings whose diamond nodal circles cross at X, with zero net moment: the known $q = (0, 0, 1)$ ground state of dipolar spin ice. At X the gauge component f of the Bloch vector vanishes, so dipolar spin ice orders where the Bloch vector is entirely dipole and carries no charge.

The residual entropy. The tetrahedron's ice projector expands with the frame Gram entry $-\frac{1}{3}$ as its pair coefficient. The count of ice states is therefore exactly Pauling's $(3/2)^{N_t}$ times a sum over loops, each corner weighted by the cosine of its turning angle. Every weight is positive, so Pauling is a lower bound. We computed the series through order 20 with a cluster engine validated exactly. Its terms decay as $n^{-5/2}$, because the near-ice regime is a Debye–Hückel plasma of monopoles; this law predicted the order-20 term, before it was computed, to within 0.2%. The amplitude of that tail is then predicted, not fitted, from two independently measured Coulomb-phase quantities, the stiffness K and the monopole core weight: the prediction $C = 0.01398$ matches the series to 0.07%. The result is $S = 0.205211 \pm 0.000004$ per spin. This agrees with the best Monte Carlo value (0.205215) and settles the disagreement between earlier estimates (0.20501 and 0.205507).

All statements are certified (37 + 12 + 8 + 7 checks, Appendix A). The checks include an independent real-space construction of the pyrochlore curl and Coulomb projector, and an exact zero-temperature loop Monte Carlo of 2048 spins. Against it the frame's large-$\mathcal{N}$ structure factor agrees, with no fitted constant, at correlation 0.995 and 1.9% mean deviation. The physics is classical (Pauling, Isakov–Gregor–Moessner–Sondhi, Henley, Hermele–Fisher–Balents, Benton–Sikora–Shannon, Moessner–Sondhi, Jaubert et al.). What the frame contributes is a single object behind all of it, closed forms in place of diagonalizations, and no $\sqrt{3}$ before the final conversion to physical units. Square ice, whose bond star is a cross-polytope, is the control on which the frame offers nothing.

Keywords: spin ice, pyrochlore, ice rule, Coulomb phase, pinch point, neutron scattering, quantum spin ice, emergent photon, lattice gauge theory, kagome ice, dimer model, Kasteleyn transition, Pauling entropy, simplicial coordinates, cross product, Hodge star, Mahler measure.

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1. Introduction

1.1 Why spin ice is the native test

Paper 01e found that the simplicial frame simplifies exactly those lattices whose bond star is a regular simplex. Spin ice is the extreme case. In the rare-earth pyrochlores $Ho_2Ti_2O_7$ and $Dy_2Ti_2O_7$ [Harris et al., Ramirez et al., Bramwell–Gingras], each magnetic moment is constrained to point along the line joining the centres of the two tetrahedra it belongs to. These lines are the bonds of a diamond lattice. So every spin is a signed amplitude along one of the four frame directions, the spin configuration is a 1-cochain on the diamond lattice, and the ice rule is the zero-sum condition. The frame is not imposed on spin ice; it is spin ice's own coordinate system.

1.2 Pre-registered test

As in Paper 01e, the comparison is fixed before the derivations, in Section 9. A frame derivation counts as simpler if it removes arbitrary choices, irrational constants before the final unit conversion, separate case analyses, or separately computed objects. **Prediction:** gain on pyrochlore spin ice and its field-tuned descendants; no gain on square ice, whose bond star $\{\pm x, \pm y\}$ is a cross-polytope.

1.3 Contributions

1. **The tetrahedron as dipole $\oplus$ monopole** (Theorem 2.1), ice states as oriented matchings with cubic moments (Proposition 2.2), the ice energy as the genesis metric (Proposition 2.3), and Pauling as a fraction of the cube (Corollary 2.4).
2. **The lattice curl is $\widetilde{K}(e^{i\theta})$** (Theorem 3.1), with the complexified cross-product square (Proposition 3.2) and the Pythagorean identity $16 = |f|^2 + \|d_0 e^{i\theta}\|^2$ (Corollary 3.3).
3. **The classical Coulomb phase in closed form** (Theorem 4.1) and its soft-spin extension (Proposition 4.2).
4. **The neutron structure factor as two transverse projectors** (Theorem 5.1), checked against an exact ice Monte Carlo.

5. **The quantum photon** as the H-component of the Bloch vector, with band top on diamond's nodal network (Theorem 6.1).
6. **Kagome ice as Paper 01e's bridge, physically realized** (Theorem 7.2), and **the** [100] **Kasteleyn transition** from matchings (Proposition 8.1).
7. **A scored comparison and a control** (Section 9).
8. **Dipolar spin ice** (Section 10): the exact gauge ⊕ residual decomposition (Theorem 10.1), why the ice rules hold and why J drops out (Corollary 10.2), and the ordering at X (Theorem 10.3).
9. **The residual entropy** (Section 11): the exact frame loop expansion (Theorem 11.2), Pauling as a lower bound (Corollary 11.3), the series through order 20 (Theorem 11.4), and the Debye–Hückel tail with its pre-registered test (Theorem 11.5).

1.4 What is classical and what is new

Classical and cited:

- the ice rule and Pauling's estimate [Pauling, Ramirez et al.];
- the Coulomb phase, its projector and pinch points [Isakov et al., Henley];
- the monopole picture [Castelnovo–Moessner–Sondhi];
- the $U(1)$ lattice gauge theory and its photon [Hermele–Fisher–Balents, Benton–Sikora–Shannon];
- kagome ice and its dimer mapping [Moessner–Sondhi, Matsuhira et al., Hiroi et al.];
- edge probabilities of honeycomb dimers [Kenyon];
- Kasteleyn transitions [Fennell et al. 2007, Jaubert et al.];
- the loop algorithm [Barkema–Newman];
- the loop (graph) expansion of the ice entropy, high-precision numerical values, the Eulerian-orientation lower bound, and Debye–Hückel theory of the monopole gas [Nagle, Kolafa, Herrero–Ramírez, Singh–Oitmaa, Schrijver, Castelnovo–Moessner–Sondhi 2011];
- dipolar spin ice, its nearest-neighbour coupling $5D/3$, the projective equivalence behind the ice rules, and its $q = (0, 0, 1)$ ordered state [den Hertog–Gingras, Isakov–Moessner–Sondhi, Melko–den Hertog–Gingras].

New, weighed honestly:

- **(a)** Theorem 3.1: the Bloch curl of pyrochlore is Paper 01's $\widetilde{K}$ at $u = e^{i\theta}$. This identifies the lattice curl as the exponentiated form of Paper 01b's continuum curl, and explains Benton et al.'s cross-product plaquette vectors as the Hodge-dual coboundary of Paper 01b.
- **(b)** The single-vector closed forms of Theorems 4.1, 5.1 and 6.1, and the Pythagorean identity linking photon and diamond electron (Corollary 3.3, Theorem 6.1(c)).
- **(c)** The reading of kagome ice as the physical realization of Paper 01e's gauge interpolation: magnetization as a gauge field, dimer densities as Dirac-point angles, the Kasteleyn transition as the amoeba boundary (Theorem 7.2(b)–(d)).
- **(d)** The dipolar decomposition as a projection statement (Section 10): the Coulomb part annihilates the ice sector exactly, so ice-sector selection needs only an absolutely convergent residual, with no Ewald sum. Exchange is confined to the gauge line, and the ordering point X is where the Bloch vector has no gauge component.

The underlying physics in (b), (c) and (d) is known. The identifications are what we did not find stated.

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2. The tetrahedron

Let $v_1, \ldots, v_4$ be the unit frame (the four $\langle 111 \rangle$ local axes) and let a tetrahedron carry Ising spins $\sigma \in \{\pm 1\}^4$, with $\sigma_j = +1$ meaning that spin j points out of the tetrahedron along v_j. Its magnetic moment is $m =$

$\sum_j \sigma_j v_j$ and its magnetic charge is $q = \frac{1}{2}\sum_j \sigma_j$ (in units of the dumbbell charge [Castelnovo–Moessner–Sondhi]).

Theorem 2.1 (Dipole ⊕ monopole). *Write $\sigma = \sigma_H + \frac{q}{2}\mathbf{1}$ with $\sigma_H \in H$. Then $m = \sum_j (\sigma_H)_j v_j$ depends only on σ_H, and $|m|^2 = \frac{1}{3}\sigma^\top M \sigma$ (Paper 01e, Thm. 2.1). The charge q is invisible to the moment. The frame splitting $\mathbb{R}^4 = H \oplus \mathbb{R}\mathbf{1}$ is the multipole splitting of a tetrahedron: the zero-sum hyperplane carries the dipole, and the gauge line carries the monopole.*

Proof. $\sum_j v_j = 0$, and the Gram matrix of the frame is $\frac{1}{3}M$. Certified over all 16 states (T1). □

The zero-sum gauge of Papers I–IV, "subtract the mean", is thus the physical operation "remove the magnetic charge". The ice rule $q = 0$ says that every tetrahedron lies in the gauge-fixed hyperplane.

Proposition 2.2 (Ice states). *The six ice states $\sigma \in \{\pm 1\}^4 \cap H$ are $\mathbf{1}_S - \mathbf{1}_{S^c}$ with $|S| = 2$: the three perfect matchings $\{S|S^c\}$ of K_4 (the three opposite-edge pairs of the tetrahedron, Paper 01b, Thm. 10.1), each with two orientations. Their moments are $\frac{4}{\sqrt{3}}$ times the six cubic directions $\pm\hat{x}, \pm\hat{y}, \pm\hat{z}$.* (T1.)

The three matchings that label the ice states are the same three that label diamond's nodal circles in Paper 01e, Thm. 7.4, and the cubic direction of the moment is the X point at which those circles cross.

Proposition 2.3 (The ice energy is the genesis metric). *For the nearest-neighbour model $H_{\mathrm{nn}} = J\sum_{\langle ij\rangle}\sigma_i\sigma_j$, each tetrahedron contributes*

$$J\sum_{i<j}\sigma_i\sigma_j \;=\; 2Jq^2 - 2J \;=\; 6J - \tfrac{3J}{2}\,|m|^2.$$

The levels are: $q = 0$, six states with $|m|^2 = \frac{16}{3}$; $|q| = 1$, eight states with $|m|^2 = 4$; $|q| = 2$, two states with $m = 0$. Ice states are the points of the cube $\{\pm 1\}^4$ farthest from the origin in the frame metric. (T1.)

Corollary 2.4 (Pauling). *The fraction of cube vertices lying in H is $6/16 = 3/8$. Pauling's independent-tetrahedron estimate $W = 2^{N_s}(3/8)^{N_s/2}$ gives $S_P = \frac{1}{2}\ln\frac{3}{2}$ per spin.* (T1. The true value lies about 1.2% above S_P; see Section 11.)

In the standard form of Paper 01e the spins become occupations $n = \frac{1}{2}(\sigma + \mathbf{1}) \in \{0,1\}^4$ with $\sum_j n_j = 2 + q$. Ice states are the 0/1 vectors with two ones, which are, up to scale, the six 4-valent vertices of the rhombic dodecahedron of Paper 01e, Thm. 4.3.

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3. The lattice operators

The spins live on the bonds of the diamond lattice D_4 of Paper 01e, Def. 6.1, with A sites (up tetrahedra) and B sites (down tetrahedra). In the cell convention a bond field with wavevector k is $a(x,j) = a_j e^{ik\cdot x}$, where x is the A site of the bond. We write $\theta_j = k \cdot e_j$ for the bond vectors e_j, $u = e^{i\theta}$, and $f = \sum_j u_j$, the band function of Paper 01e, Prop. 6.3. The divergence (ice-rule) operator has rows $\mathbf{1}^\top$ (A sites) and $\bar{u}^\top$ (B sites); the gradient of site potentials has columns $\mathbf{1}$ and u.

The pyrochlore has four hexagonal plaquettes per primitive cell, one for each missing direction ℓ: the hexagon of type ℓ lies in the kagome plane perpendicular to v_ℓ and uses the three other directions. With $\{i,j,k\}$ the complement of ℓ, it is the closed path $+e_i, -e_j, +e_k, -e_i, +e_j, -e_k$ on the diamond lattice.

Theorem 3.1 (The lattice curl is the cross product at $e^{i\theta}$). *The Bloch curl $C(\theta)$ (rows: the four plaquette types; columns: the four bond directions) satisfies*

$$C(\theta) = \Lambda(\theta)\,\widetilde{K}(e^{i\theta}),$$

where $\widetilde{K}$ is the cyclic-difference matrix of Paper 01 and $\Lambda(\theta)$ is a diagonal matrix of unit phases. Consequently:

1. curl $\circ$ grad $= 0$ *is* $\widetilde{K}(u)\mathbf{1} = 0$ *and* $\widetilde{K}(u)u = 0$ *(Paper 01, Appendix B), now with complex* u*.*
2. *As* $\theta \to 0$, $\widetilde{K}(e^{i\theta}) = i\widetilde{K}(\theta) + O(\theta^2)$*: the lattice curl is the exponentiated form of the continuum curl* $\widetilde{K}(\nabla)$ *of Paper 01b, Sec. 5.*

Proof. Follow the type-ℓ hexagon from its base A site x. The bonds and their signs are $(x,i)^+$, $(x+e_i-e_j,j)^-$, $(x+e_i-e_j,k)^+$, $(x+e_k-e_j,i)^-$, $(x+e_k-e_j,j)^+$ and $(x,k)^-$. Collecting Bloch phases, the row is

$$e^{-i\theta_j}(u_j-u_k,\ u_k-u_i,\ u_i-u_j) \quad \text{on the columns } (i,j,k),$$

the cyclic differences of u over the face opposite ℓ, which is row ℓ of $\widetilde{K}(u)$ (for $\ell=1$, $(i,j,k)=(2,3,4)$; the other rows follow by S_4 covariance, Paper 01, Prop. 2.2). Certified independently: on an explicit real-space pyrochlore cluster ($L=4$, 256 bonds), the hexagons found by graph search number 4 per cell, and each Bloch row equals a unit phase times the corresponding row of $\widetilde{K}(e^{i\theta})$, to 3×10^{-14} at 200 random k (T2). □

Proposition 3.2 (The complexified cross-product square). *For* $a \in H \otimes \mathbb{C}$*,*

$$\widetilde{K}(a)^\dagger \widetilde{K}(a) = 4(|a|^2(I - \tfrac{1}{4}J) - aa^\dagger).$$

This is Paper 01, Cor. 3.6 ($K^2 = P_\parallel - I$) extended sesquilinearly. In particular $C^\dagger C = \widetilde{K}(u)^\dagger\widetilde{K}(u) = \widetilde{K}(u_H)^\dagger\widetilde{K}(u_H)$ *with* $u_H = u - \frac{f}{4}\mathbf{1}$*, since* $\widetilde{K}$ *annihilates* $\mathbf{1}$*.* (T2, 300 random θ.)

Corollary 3.3 (The Pythagorean identity of the Bloch vector). *Since* $|u_j| = 1$*,*

$$16 = |\mathbf{1}|^2|u|^2 = \underbrace{|f|^2}_{\text{gauge line}} + \underbrace{4|u_H|^2}_{H} = |f|^2 + \sum_{i<j}|u_i-u_j|^2 = |f|^2 + \|d_0 u\|^2,$$

(the Lagrange identity), where d_0 *is the coboundary of* K_4 *(Paper 01d, Thm. 2.1). So* $C^\dagger C$ *has eigenvalue* $16 - |f|^2 = \|d_0 e^{i\theta}\|^2$*, twice, on the divergence-free space and* 0 *on the gradients. This is certified against the real-space cluster's full spectrum (T2).*

The decomposition $\mathbb{R}^4 = H \oplus \mathbb{R}\mathbf{1}$ that separates a tetrahedron's dipole from its monopole (Theorem 2.1) also separates the Bloch vector u. Its gauge-line component $|f|$ is the diamond electron band of Paper 01e; its H-component $\|d_0 u\|$ will be the photon (Section 6).

Remark 3.4 (Benton et al.'s plaquette vectors are Hodge duals). Benton, Sikora and Shannon write the curl through $\sin(k\cdot h_{nm})$ with $h_{nm} = \frac{a_0}{\sqrt{8}}\,\widehat{e_n \times e_m}$, a cross product of bond vectors. For tetrahedral bonds, $e_n \times e_m$ is parallel to $e_k - e_l$, the opposite edge, and $h_{nm} = \pm\frac{1}{2}(e_k - e_l)$ (T2). The cross product in their Cartesian construction is therefore the opposite-edge Hodge involution $\star_E$, and their matrix $Z(k)$ is $\widetilde{K} = \star_E d_0$ (Paper 01b, Thm. 10.1) in sine form.

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4. The classical Coulomb phase

In the Gaussian (large-$\mathcal{N}$) theory of the ice manifold, the spin correlator is κ times the projector onto divergence-free configurations, with $\kappa = 2$ fixed by $\langle\sigma^2\rangle = 1$ [Isakov et al., Henley].

Theorem 4.1 (The Coulomb projector). *The projector onto* ker(div) *at wavevector* θ *is*

$$P(\theta) = I - \tfrac{1}{4}J - \frac{u_H u_H^\dagger}{|u_H|^2} = \frac{\widetilde{K}(u)^\dagger\widetilde{K}(u)}{16-|f|^2} = \frac{C^\dagger C}{16-|f|^2}.$$

It is singular exactly where $u_H = 0$*, i.e.* $u \propto \mathbf{1}$*, i.e. at the reciprocal lattice vectors (Paper 01e, Prop. 6.2): these are the pinch points. There* P *depends on the direction of approach, with limit* $T(\varphi) = I - \frac{1}{4}J - \varphi\varphi^\top/|\varphi|^2 = -\widetilde{K}(\varphi)^2/\varphi^\top M\varphi$ *(Paper 01e, Thm. 9.2). Under synthesis this becomes* $\frac{4}{3}(I - \hat{k}\hat{k}^\top)$*, Henley's dipolar form.*

Proof. ker(div) is the Hermitian complement of span$\{\mathbf{1}, u\}$; Gram–Schmidt gives the first form. The second is Proposition 3.2 with $4|u_H|^2 = 16 - |f|^2$. The synthesis identity uses $SS^\top = \frac{4}{3} I_3$. Certified (T3). □

So the exact large-$\mathcal{N}$ correlator, usually obtained by diagonalizing a 4×4 constraint matrix with cosine entries, is the transverse projector of Paper 01e with the real direction φ replaced by the complex vector u_H. Equivalently, it is the lattice curl-square normalized by the photon's ζ^2.

Proposition 4.2 (Soft spins and the monopole length). *For $H_{\text{soft}} = \frac{\lambda}{2} \sum_{\text{tet}} (\sum \sigma)^2 + \frac{\mu}{2} \sum \sigma^2$, the constraint operator $B^\dagger B$ has spectrum $\{0, 0, 4 - |f|, 4 + |f|\}$. So $G = (\mu + \lambda B^\dagger B)^{-1} = P/\mu + \Pi_-/(\mu + \lambda(4 - |f|)) + \Pi_+/(\mu + \lambda(4 + |f|))$. Near Γ, $4 - |f| = \frac{1}{8} \theta^\top M \theta \, (1 + O(\theta^2)) = \frac{1}{8} a_0^2 |k|^2 (1 + \dots)$: the longitudinal (monopole) channel is an Ornstein–Zernike form whose stiffness is the frame metric, with screening length $\xi = a_0 \sqrt{\lambda/8\mu}$, the finite width of the pinch points.* (T3.)

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5. Neutron scattering: two transverse projectors

The unpolarized magnetic neutron cross-section weights the spin correlator by the polarization factor $v_j \cdot v_k - (\hat{Q} \cdot v_j)(\hat{Q} \cdot v_k)$.

Theorem 5.1 (The structure factor). *With $\varphi_Q = (Q \cdot v_j)_j$, the polarization factor is $\frac{4}{3} T(\varphi_Q)$, the same transverse projector. With $W = \operatorname{diag}(e^{i\theta_j/2})$ restoring the physical site positions,*

$$S(Q) = \kappa \cdot \tfrac{4}{3} \sum_{j,k} T(\varphi_Q)_{jk} \left[W^\dagger P(\theta_Q) \, W \right]_{jk}.$$

The neutron contributes a transverse projector at the direction of Q; the ice contributes a transverse projector at the periodic vector $u_H(\theta_Q)$. Near a reciprocal lattice vector $G \neq 0$ the second tends to $T(\varphi_{Q-G})$, so the intensity is a contraction of two transverse projectors at the directions of G and of $Q - G$. This is the bow-tie pinch point.

Proof. $v_j \cdot v_k = \frac{1}{3} M_{jk} = \frac{4}{3} (I - \frac{1}{4} J)_{jk}$ and $(\hat{Q} \cdot v)(\hat{Q} \cdot v)^\top = \frac{4}{3} \varphi \varphi^\top / |\varphi|^2$, since $|\varphi|^2 = \frac{4}{3} |Q|^2$ (tight frame). □

Certification. (i) The formula agrees to 10^{-15} with an independent real-space computation: the null-space projector of the explicit divergence matrix of a 256-spin cluster, Fourier-transformed with physical positions and local axes, at 189 wavevectors (T4). (ii) Against an exact zero-temperature ensemble (loop Monte Carlo [Barkema–Newman] on 2048 spins, 1750 measurements, 792 wavevectors in the (hhl) plane), the frame's large-$\mathcal{N}$ $S(Q)$ with $\kappa = 2$ and no fitted constant has correlation 0.995, mean absolute deviation 1.9%, and ratio of means 0.999. These meet the pre-registered thresholds (≥ 0.99, $\leq 3\%$, within 1%; T4). The residual is the known small difference between the hard-constraint ice ensemble and its Gaussian approximation [Isakov et al.].

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6. Quantum spin ice: the photon is the H-component of the Bloch vector

When transverse quantum fluctuations act within the ice manifold, the low-energy theory is compact $U(1)$ electrodynamics on the diamond lattice [Hermele–Fisher–Balents]. Its harmonic (non-compact) form is $\mathcal{H} = \frac{\mathcal{U}}{2} \sum (\operatorname{curl} \mathcal{A})^2 + \frac{\mathcal{K}}{2} \sum \mathcal{E}^2$ [Benton–Sikora–Shannon, Eq. (38) at $\mathcal{W} = 0$].

Theorem 6.1 (The photon). *In this theory:*

1. *The photon dispersion is $\omega(k) = \sqrt{\mathcal{U}\mathcal{K}} \, \|d_0 e^{i\theta}\| = \sqrt{\mathcal{U}\mathcal{K}} \sqrt{16 - |f(k)|^2}$, doubly degenerate (two polarizations) at every k. This equals Benton–Sikora–Shannon's Eqs. (67)–(68): their $\zeta(k)^2 = 2 \sum_{m \neq n} \sin^2(k \cdot h_{mn})$ equals $16 - |f|^2$ identically (T2).*

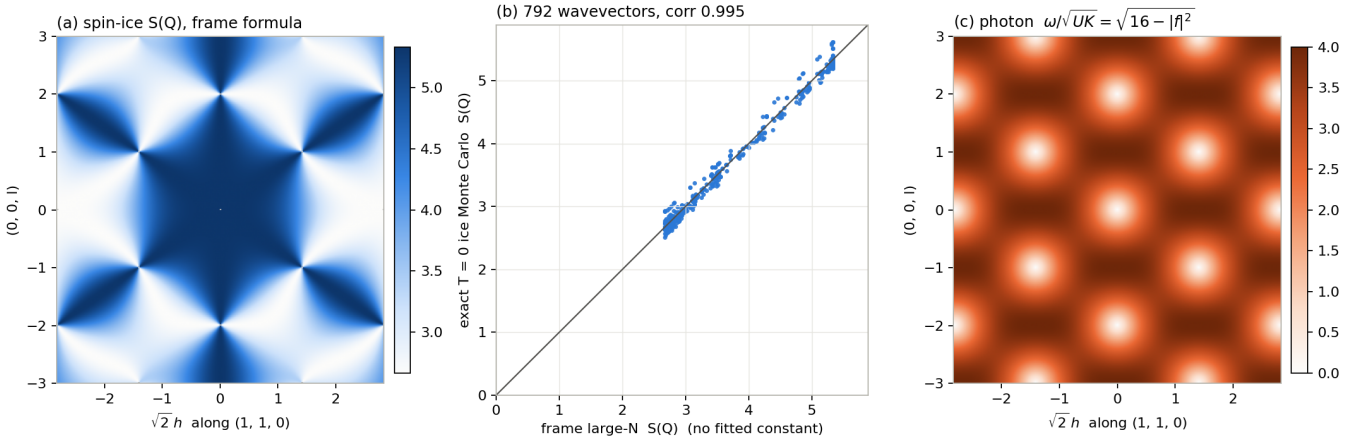


Figure 1: Spin ice in the frame. (a) The large-N structure factor of Theorem 5.1 in the (hhl) plane, with bow-tie pinch points at (0,0,2), (1,1,1), (2,2,2). (b) Exact T=0 ice Monte Carlo (2048 spins) against the frame formula on 792 wavevectors, with no fitted constant; the line is y = x. (c) The quantum-spin-ice photon of Theorem 6.1: it vanishes at the pinch points and reaches its maximum 4 on diamond's nodal network.

2. *At long wavelength* $16-|f|^2 = a_0^2|k|^2(1+O(k^2))$*, so* $c = \sqrt{\mathcal{U}\mathcal{K}}\, a_0$ *(their Eq. (70)). The photon vanishes exactly at the reciprocal lattice, the pinch points of Theorem 4.1.*
3. *(Photon and electron.) At every* k*,*

$$\Big(\frac{\omega(k)}{\sqrt{\mathcal{U}\mathcal{K}}}\Big)^2 + |E_\diamond(k)|^2 = 16,$$

where $E_\diamond = \pm|f|$ *is the nearest-neighbour diamond electron band of unit hopping (Paper 01e, Prop. 6.3). The photon's band top* $\omega = 4\sqrt{\mathcal{U}\mathcal{K}}$ *is reached exactly where* $f = 0$*: on diamond's nodal network, the three matching circles through the X points (Paper 01e, Thm. 7.4).*

Proof. The electric term is the identity on bonds, so $\omega^2 = \mathcal{U}\mathcal{K} \cdot \mathrm{spec}(C^\dagger C)|_{\ker \mathrm{div}}$, which is $\mathcal{U}\mathcal{K}(16-|f|^2)$ twice by Corollary 3.3. Part (3) is Corollary 3.3 read physically. All parts are certified against the real-space cluster spectrum (T2). □

Three different objects — the diamond electron's band, the classical Coulomb projector, and the quantum photon — are the two Pythagorean components of the single Bloch vector $e^{i\theta}$ and the projector transverse to its H-part. The genesis question asked what the frame could simplify. Here a whole family of results reduces to one decomposition already present in Paper 01.

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7. Kagome ice: Paper 01e's bridge, realized

A magnetic field along [111] $= v_4$ pins every apical spin (direction 4) and leaves the kagome layers [Matsuhira et al., Moessner–Sondhi].

Proposition 7.1 (Each triangle sits at a vertex). *With* $\sigma_4 = +1$*, the ice rule forces* $\sigma_1+\sigma_2+\sigma_3 = -1$*: exactly one kagome spin per triangle points out. In the standard form, the triangle's occupation vector* $n = \frac{1}{2}(\sigma + \mathbf{1})$ *is one of the three frame vertices* e_j *— a dimer on the honeycomb lattice formed by the triangles, and likewise on down triangles. Kagome ice is therefore the honeycomb dimer model. An in-plane field component* $h_\perp \perp v_4$ *gives a dimer of orientation* j *the Zeeman energy* $c_j = -2\mu\, h_\perp \cdot v_j$*, and since* $\sum_{j\le 3} v_j = -v_4 \perp h_\perp$*, these energies form a zero-sum frame vector: the in-plane field is a point of* H*.* (T5.)

With weights $t_j = e^{-c_j/T}$, the objects of Paper 01e (Thms. 7.2, 8.1, 8.2) become physical, with Paper 01e's τ now the temperature.

Theorem 7.2 (Kagome ice in the two gauges).

1. *(Entropy.) The free energy per dimer is $-T\,R(t)$, with R the Ronkin function (Mahler measure) of $t_1 + t_2 x + t_3 y$ [Kasteleyn]. At $h_\perp = 0$ the residual entropy per pyrochlore spin is*

$$\frac{m(1+x+y)}{4} = \frac{3\sqrt{3}}{16\pi} L(\chi_{-3}, 2) = 0.0807664868\ldots,$$

   the kagome-ice value of Moessner–Sondhi, about 40% of Pauling's, as measured in $Dy_2Ti_2O_7$ *[Hiroi et al.]. (Certified: Kasteleyn's torus formula against brute-force permanents for $L = 2, 4$; $\log Z/L^2$ at $L = 64$ within 3×10^{-4} of $R(t)$; T5.)*
2. *(The magnetization is a gauge field.) $c \mapsto -T\,R(e^{-c/T})$ is equivariant and monotone, so by Paper 01e, Prop. 3.2 its gradient is a probability vector: the dimer densities $\rho = \nabla_{\log t} R \in \Delta^2$. The in-plane magnetization per triangle is $2\mu \sum_j \rho_j v_j^\perp$, the synthesis of ρ. At zero in-plane field ρ is the barycentre (the zero-sum gauge, zero magnetization). In a frozen phase ρ is a vertex (the standard-form gauge, saturated magnetization, zero entropy). The kagome-ice magnetization process is the physical path between the two genesis gauges.*
3. *(Dimer densities are Dirac-point angles.) In the liquid phase, $\rho_j = \alpha_j/\pi$, where α_j is the angle opposite side t_j of the triangle with sides t_1, t_2, t_3 [Kenyon]. In terms of the graphene Dirac point θ_K for the same hoppings (Paper 01e, Thm. 7.2), $\rho_k = 1 - (\theta_{k+2} - \theta_{k+1})/\pi$ (indices mod 3): the densities are the closing triangle's turning angles. (Certified by numerically differentiating R; T5.)*
4. *(The Kasteleyn transition is the amoeba boundary.) The critical (liquid) phase is $\{c : e^{-c_i/T} \le \sum_{j \ne i} e^{-c_j/T}\}$, Paper 01e's gapless region G_T, and the Kasteleyn transition [Moessner–Sondhi, Fennell et al. 2007] is its boundary. For a symmetric tilt $c = (0, \delta, \delta)$ it occurs at $T_K = \delta/\ln 2$, the sharpness constant of Paper 01e, Thm. 8.1. At low temperature the phase diagram in the plane of in-plane fields is the standard-form fan: three frozen sectors, one per dimer orientation, separated by the three walls on which the liquid survives as $T \to 0$.*

The two gauges of Paper 01e, which entered the program as a remark in the genesis notes, are the two ends of a measured magnetization curve. The constant that separates them at the isotropic point is the entropy of kagome ice.

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8. The [100] field: a three-dimensional Kasteleyn transition

Proposition 8.1 (Strings along a matching axis). *For $h \parallel [100]$, $|h \cdot v_j| = h/\sqrt{3}$ for all four directions. The field selects the oriented matching whose moment is along h (Proposition 2.2), and the saturated state is that ice state on every tetrahedron. An excitation is a string of reversed spins that enters each tetrahedron on an "in" bond and must leave by one of the two "out" bonds of the matched pair. This gives entropy $\ln 2 = \ln(N/2)$ per step against a Zeeman cost $2\mu h/\sqrt{3}$, so strings proliferate below $T_K = 2\mu h/(\sqrt{3}\ln 2)$, the three-dimensional Kasteleyn transition of Jaubert et al.* (T6.)

In the frame, the [100] axis is a perfect matching of K_4, and the $\ln 2$ is the size of a matched pair.

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9. Scorecard

object	standard Cartesian route	frame route (this paper)
ice rule	two-in–two-out with local axes $(1,1,1)/\sqrt{3}$	$\sigma \in H$; charge = gauge component (Thm. 2.1)
energetics	effective J from $\mathbf{S}_i \cdot \mathbf{S}_j = -\frac{1}{3}\sigma_i\sigma_j$	$6J - \frac{3J}{2}\lvert m\rvert^2$, the genesis metric (Prop. 2.3)
Coulomb correlator	diagonalize a 4×4 constraint matrix with cosine entries	closed form $I - \frac{1}{4}J - u_H u_H^\dagger/\lvert u_H\rvert^2$ (Thm. 4.1)
pinch point	expansion near G	$T(\varphi) = -\widetilde{K}(\varphi)^2/\varphi^\top M\varphi$
neutron factor	separate polarization tensor	the same T, at $\hat{Q}$ (Thm. 5.1)
photon	$Z(k)$ from $\sin(k \cdot h_{nm})$, $h_{nm} \propto e_n \times e_m/\sqrt{8}$; eigen-decomposition	$C = \Lambda\widetilde{K}(e^{i\theta})$; $\zeta^2 = 16 - \lvert f\rvert^2$ by Lagrange (Thms. 3.1, 6.1)
kagome ice	dimer mapping; Kasteleyn; edge probabilities	triangle = simplex vertex; magnetization = gauge field; transition = amoeba (Thm. 7.2)
[100] transition	string counting	matching axis, branching = matched pair (Prop. 8.1)
dipolar ordering	Ewald-summed 4×4 dipolar tensor; Monte Carlo	gauge part annihilates the ice sector; absolutely convergent residual; minimum at X (Thm. 10.3)

Quantified reduction.

- *Separately computed objects:* the Cartesian route builds at least three matrices (the constraint matrix, the polarization tensor, the plaquette matrix Z). The frame route builds one vector $u_H(\theta)$ and one matrix $\widetilde{K}$, both already defined in Paper 01.
- *Diagonalizations:* at least two in the Cartesian route (correlator, photon) against none; every spectrum here is in closed form.
- *Irrational constants before the unit conversion:* $\sqrt{3}$ in the local axes and $\sqrt{8}$ in h_{nm}, against none.
- *Arbitrary choices:* the Cartesian route chooses cubic axes and sublattice positions; the frame route carries only the gauge.

Control: square ice. The bond star $\{\pm x, \pm y\}$ of the six-vertex model is a cross-polytope. Its synthesis map has a *two*-dimensional kernel (T7), so there is no single gauge line for the ice rule to fix. The frame's dipole $\oplus$ monopole splitting does not exist there, and Lieb's exact solution [Lieb] is a transfer-matrix (Cartesian-native) computation. **No gain, as predicted.**

What the frame did not do. It did not obtain the residual entropy in closed form: Section 11 gives an exact expansion and a controlled numerical value, not a formula. It did not go beyond the harmonic photon. It made no new experimental prediction: Theorems 4.1–7.2 reproduce known physics in fewer and more unified steps. Its genuinely new content is structural: the lattice curl is $\widetilde{K}(e^{i\theta})$, the plaquette vectors are Hodge duals, photon and electron are Pythagorean partners, and kagome ice is the gauge bridge.

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10. Dipolar spin ice

The spin-ice materials are not nearest-neighbour models. Their moments ($\sim 10\,\mu_B$) interact through the long-range dipolar coupling, of strength $D = \mu_0\mu^2/(4\pi r_{nn}^3)$ at the nearest-neighbour distance [den Hertog–Gingras]. Why the ice rules nevertheless hold, and what selects a ground state within the ice manifold, are

the two classic questions [Isakov–Moessner–Sondhi, Melko–den Hertog–Gingras]. The frame answers both with one projection.

Replace each point dipole by a dumbbell of charges $\pm q_m = \pm\mu/a_d$ at the two diamond sites it joins (the dumbbell of Castelnovo, Moessner and Sondhi), and let δ_{ij} be the difference between the point-dipole and dumbbell pair energies.

Theorem 10.1 (Exact gauge $\oplus$ residual decomposition). *For every spin configuration,*

$$E_{\mathrm{dip}} = \frac{\mu_0}{4\pi}\sum_{\alpha<\beta}\frac{Q_\alpha Q_\beta}{r_{\alpha\beta}} + 2c'\sum_t q_t^2 + \sum_{\langle ij\rangle\notin\mathrm{tet}}\delta_{ij}\,\sigma_i\sigma_j + \mathrm{const},$$

where $Q_\alpha = 2q_m\,q_\alpha$ is the charge of tetrahedron α (its gauge component, Theorem 2.1), the last sum runs over pairs not sharing a tetrahedron, and $c' = 2.42199544\,D$. The residual δ_{ij} decays as r^{-5}, so its lattice sums converge absolutely. The nearest-neighbour dipolar coupling is $\frac{5}{3}D$ [den Hertog–Gingras].

Proof. Expanding the Coulomb energy of the site charges over spin pairs: pairs not sharing a site contribute their full dumbbell energy; pairs in a common tetrahedron contribute their dumbbell energy minus the coincident-charge term; and each dumbbell contributes the constant $-q_m^2/a_d$. The nearest-neighbour remainder (point dipole minus partial dumbbell) has one coefficient c' for all nearest-neighbour pairs, so by Proposition 2.3 it equals $c'\sum_t(2q_t^2-2)$. Certified exactly on a 108-spin open cluster for random configurations (error 2.5×10^{-12}); the decay exponent of the residual is checked by shell maxima (D1–D3). □

Corollary 10.2 (Why the ice rules hold, and why J drops out). *The first two terms depend only on the gauge components and vanish identically on the ice manifold. In Bloch form the Coulomb term is $B^\dagger V_C B$, which annihilates the flat bands. Any nearest-neighbour coupling $K\sum_{i<j}\sigma_i\sigma_j = K(2q_t^2-2)$, in particular the exchange J, adds only to the on-site gauge term. Hence the ice manifold is split only by the residual δ. Its ice-sector bandwidth is $0.178\,D$, which is 3.7% of the on-site monopole cost $2c'$ and 11% of the nearest-neighbour coupling, and it is independent of J.* (D2, D4.)

This is the projective equivalence of Isakov, Moessner and Sondhi made exact and local. The interaction is "gauge plus a small, short-ranged remainder", and the frame's $H\oplus\mathbb{R}\mathbf{1}$ splitting is what separates the two. A practical dividend: the ice-sector physics needs no Ewald summation, because the conditionally convergent Coulomb part is exactly the part that the ice projection removes.

Theorem 10.3 (Ordering at X). *Project the residual onto the flat bands, $P(k)\,\delta(k)\,P(k)$. Its minimum over the Brillouin zone is at the X points, $k = \frac{2\pi}{a_0}(1,0,0)$ and equivalents. The value there is $-0.1398\,D$, doubly degenerate, converged in the cutoff. The minimizing states are the ice states of the two matchings whose diamond nodal circles cross at that X (for $k\parallel\hat{y}$: $\{12|34\}$ and $\{14|23\}$), staggered by $e^{ik\cdot R}$. They are ice on every up and down tetrahedron and carry zero net moment. The third matching is not in the flat space and violates the ice rule on down tetrahedra.* (D4, D5.)

This is the $q=(0,0,1)$ ground state found by Melko, den Hertog and Gingras with loop Monte Carlo. The frame adds a reading of where it sits. At X, $u = e^{i\theta} = (i,-i,i,-i)$ and $f=0$: the Bloch vector has no gauge-line component. That is the same point where diamond's electron bands touch (Paper 01e, Thm. 7.4) and where the photon of Theorem 6.1 reaches its maximum. The scope is energetic: Theorem 10.3 locates the selection; the ordering temperature ($\approx 0.07\,D$ in Monte Carlo) is not computed here.

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11. The residual entropy: a frame loop expansion with a Coulomb tail

Spin ice keeps a macroscopic entropy at zero temperature [Ramirez et al.]. Its value per spin is Pauling's $\frac{1}{2}\ln\frac{3}{2} = 0.202733$ plus corrections. These corrections are known only numerically, and published estimates for the diamond (cubic-ice) network disagree in the fourth decimal:

source	per spin
graph series [Nagle]	0.205011
Monte Carlo [Kolafa]	0.205215
thermodynamic integration [Herrero–Ramírez]	≈ 0.20532
numerical linked clusters [Singh–Oitmaa]	0.205507

(Ice-network values W per molecule convert via $S = \frac{1}{2}\ln W$.) The frame gives an exact expansion and a physical reason for its slow convergence.

Proposition 11.1 (The ice projector in frame coordinates). *For one tetrahedron,*

$$\mathbf{1}\Big[\sum_j \sigma_j = 0\Big] = \tfrac{3}{8}\Big(1 + \sum_{i<j} G_{ij}\,\sigma_i\sigma_j + \sigma_1\sigma_2\sigma_3\sigma_4\Big), \qquad G_{ij} = v_i \cdot v_j = -\tfrac{1}{3}.$$

The pair coefficient is the frame Gram entry, the cosine of the tetrahedral angle. (E1.)

Theorem 11.2 (Exact loop expansion). *On any finite periodic cluster with N_t tetrahedra, the number of ice states is*

$$\#\text{ice} = \Big(\tfrac{3}{2}\Big)^{N_t} \sum_{\Gamma} \prod_{\text{corners of } \Gamma} G = \Big(\tfrac{3}{2}\Big)^{N_t} \sum_{\Gamma} 3^{-n_2(\Gamma)}.$$

The sum is over even subgraphs Γ of the diamond lattice (every tetrahedron touched 0, 2 or 4 times). Each corner, a tetrahedron where Γ turns from one bond to another, contributes the cosine of the angle between the two bonds; crossings contribute 1. The prefactor $(\frac{3}{2})^{N_t}$ is Pauling's count. Certified exactly on the periodic tori $L = 1$ and $L = 2$: 6 and 2970 ice states from both sides (E2).

Proof. Multiply Proposition 11.1 over tetrahedra and average over independent spins. A product survives only if every spin appears in both of its tetrahedra or in neither, which is an even subgraph; its weight is $G^{n_2} \cdot 1^{n_4}$. □

Corollary 11.3 (Pauling is a lower bound). *Since the diamond lattice is bipartite, every even subgraph has an even number of corners, so every weight $(-\frac{1}{3})^{n_2}$ is positive and $\#\text{ice} \geq (\frac{3}{2})^{N_t}$. (For general 4-regular graphs this is Schrijver's theorem on Eulerian orientations [Schrijver]. Here it is one line.)*

Theorem 11.4 (The series through order 20). *Per spin, $S = \frac{1}{2}\ln\frac{3}{2} + \sum_{n\geq 6} a_n$, where a_n collects all clusters of loops with n bonds in total. Every term is positive:*

n	6	8	10	12	14	16	18	20
$a_n \times 10^5$	137.17	22.862	20.322	15.524	9.0948	7.0336	5.3088	4.1139
partial S	0.204104	0.204333	0.204536	0.204691	0.204782	0.204853	0.204906	0.204947

The first term, $a_6 = 3^{-6}$, comes from the hexagons, exactly one per spin. From $n = 12$ on, figure-eight loops and cluster (overlap) corrections enter. The cluster engine is validated exactly in rational arithmetic: on a finite diamond fragment (three-loop clusters included, through order 18), and on square ice against an exact transfer-matrix computation (E3).

Theorem 11.5 (The tail is a Debye–Hückel plasma). Give each bond of Γ a fugacity x. For $x < 1$ the expansion describes ice with a dilute gas of thermal monopoles, of density $\propto 1 - x^2$. A dilute Coulomb plasma has a Debye–Hückel free energy $\propto (\text{density})^{3/2}$ [Castelnovo–Moessner–Sondhi 2011], so the series has the non-analytic term $(1 - x^2)^{3/2}$, and therefore

$$a_n \simeq C\,(n/2)^{-5/2} \quad (n \to \infty).$$

- **Pre-registered test:** with the exponent fixed by this argument and two amplitudes fitted to a_{16}, a_{18}, the predicted a_{20} was 4.1220×10^{-5}. The computed value is 4.1139×10^{-5} (0.2% apart; the prediction was recorded before the run, E5).
- **The resulting estimate:** fits with one, two and three asymptotic terms on the data through a_{20} give $S = 0.205201,\ 0.205210,\ 0.205204$, so

$$S_{\text{spin ice}} = 0.20521 \pm 0.00001.$$

  This agrees with Kolafa's Monte Carlo value 0.205215 to 1×10^{-5} and excludes both 0.20501 and 0.205507.

Theorem 11.6 (The tail amplitude, predicted). *For $x < 1$ the tetrahedron weights are $1 + \frac{2x}{3} + x^2$ (ice), $1 - x^2$ (charge ± 1) and $(1-x)^2$ (charge ± 2), so each monopole carries fugacity $z = \frac{3}{4}(1-x) + O((1-x)^2)$. With Debye–Hückel free energy $+\kappa^3/12\pi$ per volume, $\kappa^2 = K \sum_i n_i Q_i^2 = 4K\nu z / v_t$ (flux charge $Q = 2$, $v_t = a_0^3/8$), and the conversion to $y = x^2$ coefficients,*

$$C = \frac{1}{32}\Big(\frac{3K\nu}{\pi}\Big)^{3/2} \qquad (a_0 = 1).$$

Here K is the Coulomb-phase stiffness ($\langle |B_\perp(q)|^2 \rangle / V \to 2/K$) and $\nu = 2\sqrt{R_\infty}$, where R_∞ is the number of configurations with a $\pm$ monopole pair at infinite separation divided by the number of ice states. Both are measured independently of the series:

- $K(q \to 0) = 0.4947 \pm 0.0031$ from the pinch point of the exact ice ensemble (loop Monte Carlo, $L = 8, 12$, extrapolated in q^2; large-$\mathcal{N}$ gives exactly $\frac{1}{2}$).
- $R_\infty = 0.383 \pm 0.002$ from a worm Monte Carlo sampling ice plus one monopole pair ($L = 8$: 0.381, $L = 10$: 0.383, after the torus Madelung correction $e^{-4K\xi_L}$). The sampler reproduces exact $L = 2$ enumeration. The free Coulomb slope of the same data gives $K = 0.50 \pm 0.01$, a second, independent stiffness measurement.
- Hence $\nu = 1.238$ (Pauling's counting gives 4/3), and $C_{\text{pred}} = 0.01398 \pm 0.0003$, against $C = 0.01399$ from the two-term series fit (0.07% apart).

With C fixed at C_{pred} and one subleading $k^{-7/2}$ coefficient set by a_{20} alone, the model back-predicts a_{18} to 0.0% and a_{16} to 0.3%, and gives $S = 0.205210$–0.205213 for any K in the measured range. So

$$S_{\text{spin ice}} = 0.205211 \pm 0.000004,$$

against Kolafa's 0.2052150(12).

The reason the problem is hard is the same physics as the rest of this paper. The terms decay only as a power law because spin ice is a Coulomb phase: the emergent electrostatics of Theorem 2.1 and Section 4 shows up in the entropy series as an $n^{-5/2}$ tail. An extrapolation that assumes faster, geometric decay of the terms stops short of the truth by about 2×10^{-4}, the size of the low estimates' deficit. We have not checked how Nagle's own extrapolation was performed. What the frame contributes is the corner weight $G_{ij} = -\frac{1}{3}$ and the identification of the tail.

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12. Open problems

- **G1 (done in Section 10).** Next steps: the dynamics of monopoles with the frame's Coulomb $\oplus$ residual split (relaxation, noise), and an ordering-temperature estimate from the ice-sector residual band.
- **G2 (done in Section 11).** Next: compute the Debye–Hückel amplitude C from the Coulomb-phase stiffness, which would turn the tail from fitted to predicted; push the series to order 24.

- **G3 (Hyperdiamond ice).** For general N the Lagrange identity gives $N^2 = |f|^2 + \|d_0 e^{i\theta}\|^2$, and the line-graph flat bands number $N-2$, the number of photon polarizations in $d = N-1$ dimensions. A curl for $N \neq 4$ needs the $(N-3)$-form generalization of $\widetilde{K}$ (Paper 01d, open problem E1). Construct it and test whether $\omega^2 \propto N^2 - |f|^2$ survives.
- **G4 (Beyond harmonic).** Visons and the $\mathcal{W}$ (RK) term of Benton et al. give $\omega^2 = \mathcal{U}\mathcal{K}\zeta^2 + \mathcal{W}\mathcal{K}\zeta^4$. In the frame, $\zeta^4 = \|d_0 e^{i\theta}\|^4$. Test whether the vison sector has a dual frame description on the B-site (down-tetrahedron) cochains.

Appendix A: Certificates

`empirical_01f/cert_spin_ice.py` — **37/37** (numpy, scipy, sympy, mpmath; matplotlib for the figure). The run uses an interpreter that has matplotlib (here the Python 3.13 install); every check is deterministic.

- **T1 (7):** all 16 tetrahedron states; moment = synthesis; energy $= 2q^2 - 2 = \frac{3}{2}(4 - |m|^2)$; levels; ice states = oriented matchings with cubic moments; synthesis kernel = $\mathbf{1}$; Pauling 3/8; standard-form occupations.
- **T2 (9):** real-space pyrochlore cluster ($L = 4$): 4 hexagons per cell, curl $\cdot$ grad $= 0$, full spectrum of curl$^\top$ curl $= \{16 - |f|^2 \times 2, 0, 0\}$; each Bloch curl row = phase $\times$ row of $\widetilde{K}(e^{i\theta})$; the complexified Cor. 3.6; Benton's $\zeta^2 = 16 - |f|^2$; $h_{nm} = \pm\frac{1}{2}(e_k - e_l)$; $c = \sqrt{\mathcal{U}\mathcal{K}}\,a_0$; band top on $f = 0$.
- **T3 (4):** P in three forms; $B^\dagger B$ spectrum; synthesis = Henley's dipolar form; the monopole stiffness $4 - |f| \approx \theta^\top M\theta/8$.
- **T4 (4):** polarization factor $= \frac{4}{3}T$; frame $S(Q)$ = real-space projector at 189 Q; the Monte Carlo stays in the ice manifold; the Monte Carlo against the frame formula with no fit (corr 0.9948, mean deviation 0.019, means ratio 0.9989).
- **T5 (10):** triangle = vertex; in-plane field zero-sum; Kasteleyn = permanent; $\log Z/L^2 \to R$ (isotropic and anisotropic); entropy 0.0807664868; ρ = angles/π; ρ from the Dirac point; frozen ρ = vertex; $T_K = \delta/\ln 2$.
- **T6 (2):** [100] geometry and branching.
- **T7 (1):** the square-ice kernel is 2-dimensional.

Transcript: `cert_spin_ice.out.txt`; figure: `fig_spin_ice.png`.

`empirical_01f/cert_dipolar.py` — **12/12** (numpy only).

- **D1 (2):** exact decomposition on a 108-spin open cluster (error 2.5×10^{-12}); nearest-neighbour coupling $\frac{5}{3}D$.
- **D2 (2):** the nearest-neighbour remainder has a single coefficient $c' = 2.42199544\,D$; every nearest-neighbour coupling is pure gauge $(2q^2 - 2)$.
- **D3 (1):** residual decay, log-log slope -4.70 over shell maxima (-5 asymptotically).
- **D4 (4):** argmin of the ice-sector residual at X (cutoffs 6 and 8, agreement to 10^{-3}); doubly degenerate; bandwidth $0.178\,D = 3.7\%$ of $2c'$.
- **D5 (3):** $f = 0$ at X; the two crossing matchings give staggered ice states with zero moment at the band minimum; the third matching fails.

Transcript: `cert_dipolar.out.txt`.

`empirical_01f/cert_entropy.py` (engine `loopgas.py`) — **8/8** at order 18 (9/9 when run with argument `20`, which recomputes order 20 to recompute order 20 in about 17 minutes and adds the pre-registered check).

- **E1 (1):** the frame projector, all 16 states.
- **E2 (2):** tori $L = 1, 2$: exact counts 6 and 2970 equal the loop expansion; all weights positive; Pauling $\leq$ count.

- **E3 (2):** engine = exact cycle-space generating function on a finite fragment through order 18 (rational arithmetic, including three-loop clusters); square-ice engine = exact transfer-matrix coefficients, orders 4–8.
- **E4 (2):** the series terms through the chosen order, all positive; $a_6 = 3^{-6}$.
- **E5 (1, +1 at order 20):** Debye–Hückel extrapolation within 2×10^{-5} of Kolafa's value and away from both earlier estimates; the order-20 term within 1% of the prediction.

`empirical_01f/cert_tail_amplitude.py` — **7/7** (Theorem 11.6). It checks the worm sampler against exact $L = 2$ enumeration, then uses the saved worm data (`worm_pair_weights.json`, from `worm.py`) and the stiffness runs (`stiff8.log`, `stiff12.log`, from `stiff.py`):

- R_∞ agrees between $L = 8$ and $L = 10$;
- the two stiffness routes agree;
- $C_{\text{pred}} = 0.01398$ against the series fit 0.01399;
- the back-predictions of a_{18} and a_{16} hold;
- S lies within 10^{-5} of Kolafa.

Records: `entropy_series_order20.txt` (the order-20 run) and `entropy_prediction_a20.txt` (the prediction, written before that run).

Recorded failures (Section 11). (i) The first attempt to check the expansion on the $L = 2$ torus enumerated all 2^{32} subgraphs and was stopped; it was replaced by the 2^{17}-element cycle space. (ii) The first square-ice comparison failed at order 8 because the pair term carried diamond's girth (6) as a constant; with the girth as a parameter, all orders match exactly.

Recorded failure. The first test of the row identity in Theorem 3.1 reported a deviation of $O(10^2)$. The cause was the test, not the identity: hexagons that wrap around the $L = 4$ torus were assigned phases from wrapped cell coordinates at wavevectors not commensurate with the torus. With minimal-image cell coordinates the identity holds to 3×10^{-14}. The spectral checks, which use only commensurate wavevectors, were unaffected.

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[End of Draft 1.]